

Does Compositional Refinement Strictly Extend Heuristic Specification Inference? A Measurement Study

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Abstract—Heuristic specification inferers compute a per-method specification from local syntactic evidence. When a method calls another, the callee’s specification is typically incorporated by single-pass propagation: at each call site, the callee’s preconditions (substituted for actual arguments) are appended to the caller’s precondition set. A natural question is whether this single pass is *compositional* in the sense that matters — whether a second, top-down weakest-precondition (WP) refinement pass would discover additional preconditions that the isolated pass missed. We implement the second pass for an existing JML inferer (a 414-line analyser that walks each method’s body in reverse-topological order, lifts callee specifications through enclosing branch guards, emits disjunctive obligations over polymorphic-dispatch candidates, and iterates to a fixpoint on cyclic strongly-connected components) and measure its effect on five open-source Java libraries spanning four idiom families (Apache Commons Lang, Commons IO, Commons Math, jOOL, Vavr): 1,712 source files and 21,052 inferred methods in total. The result is unambiguous: the second pass refines 23–43% of methods in each library’s cache (median 35.53%) and adds 45,740 new precondition clauses overall. Branch-conditional implications of the form `guard ==> obligation` dominate on four libraries of five (46–67% of new clauses), and disjunctive obligations from polymorphic dispatch are present on every library (1.8–9.8%); both are shapes the single-pass propagation cannot produce because it lacks the control-flow context the WP walk supplies and resolves each call to a single nominal callee. The single pass does *not* subsume the compositional pass on any library studied. A version-step instance — inferring across Commons Lang 3.12.0 and 3.14.0 — surfaces 525 candidate behavioural changes and 144 strengthened preconditions, the candidate-breaking artefact a specification-level compatibility checker would consume.

Index Terms—JML, specification inference, compositional reasoning, weakest precondition, Java, static analysis

I. INTRODUCTION

A heuristic specification inferer for Java produces, for each method, a set of Java Modeling Language (JML) clauses inferred from local syntactic evidence: a null-dereference of a parameter inside the body becomes a `requires \p != null` clause; an array read at index `i` becomes `requires 0 <= i && i < a.length`; and so on. Recent work has shown that such specifications, even when heuristically produced, materially improve downstream test generation [1] and can be embedded in compiled artefacts with full lossless

roundtrip [2]. Both of those results depend on the spec set being *rich enough* to constrain downstream tools — the same dependency motivates work on large-language-model (LLM) driven test oracle generation [3]–[6], LLM-driven contract synthesis [7]–[10], and dynamic-trace invariant detectors such as Daikon [11].

The dominant strategy for incorporating cross-method information in such an inferer is *single-pass propagation*: while inferring method `m`, the analyser looks up each callee’s cached specification, substitutes the actual arguments at the call site, and appends the substituted callee preconditions to `m`’s precondition set. This is fast and order-independent (provided the callee is inferred first), and it is what the inferer of [1] already does via its `InterproceduralAnalyzer`. A natural question is whether this single pass captures everything a *genuinely compositional* analysis would: that is, whether a second, top-down weakest-precondition (WP) refinement pass [12]–[14], walking each method’s body with the callee specifications substituted at each call site, would discover any additional clauses that the single pass missed. Compositional procedure summaries are foundational in static analysis [15], [16] and underpin frameworks such as Infer [17] and Houdini [18], but those frameworks operate on different intermediate representations and do not produce JML clauses directly.

The question matters operationally. If the answer is no, the second-pass refactoring (and the associated implementation, testing, and maintenance cost) is unjustified; the simpler single-pass architecture is good enough. If the answer is yes, downstream consumers of inferred specifications — test generators, verifiers, LLM-based tooling — see a measurably richer spec set when the second pass runs.

In this paper, we implement the second pass and answer the question empirically on real Java code. We address the following research questions:

- **RQ1.** Does a top-down WP-style compositional refinement pass strictly extend the precondition set produced by an established single-pass propagation, or is the single pass already sufficient?
- **RQ2.** If the second pass adds clauses, which structural shapes (null checks, range bounds, branch-conditional implications, disjunctive obligations) dominate the additions?

Replication package (analyser source, harness, per-library logs, parsed clause samples) at <https://anonymous.4open.science/r/jml-inferer-anon> during review; non-anonymous DOI-archived release in the camera-ready version.

The answer characterises *why* the single pass falls short.

- **RQ3.** What is the runtime cost of the second pass at corpus scale, and how does it interact with cyclic strongly-connected components (mutual recursion) in the call graph?

We answer RQ1–RQ3 on a five-library corpus spanning four idiom families: Apache Commons Lang 3.14.0 (3,494 methods), Apache Commons IO 2.13.0 (2,111 methods), Apache Commons Math 3.6.1 (6,645 methods), jOOL 0.9.14 (3,691 methods), and Vavr 0.10.4 (5,111 methods); combined, 21,052 inferred methods over 1,712 source files. Lang and IO are the same libraries used by the predecessor work [1], [2]. The principal findings are summarised in Table I and analysed in Section V; in short, the second pass refines 23–43% of methods in each library’s cache (median 35.53%) and adds 45,740 new precondition clauses overall, with branch-conditional implications (`guard ==> obligation`) the dominant shape on four libraries out of five and disjunctive clauses (the polymorphic-dispatch signal) present on every library.

Contributions. In summary, this paper makes the following contributions:

- A 414-line `CompositionalAnalyzer` that implements the top-down WP refinement pass over a populated specification cache, supporting per-call-site argument substitution, enclosing-guard lifting, polymorphic-dispatch disjunction over candidate callees, and bounded fixpoint iteration on cyclic strongly-connected components.
- An empirical measurement on five real-world Java libraries spanning four idiom families (21,052 inferred methods in total) showing that the second pass refines 23–43% of methods in each library’s cache and adds 45,740 new precondition clauses overall; the single-pass strategy is *not* subsuming on any library.
- A shape-categorisation of the newly-added preconditions showing that branch-conditional implications (`guard ==> obligation`) dominate on four of five libraries (46–67% of new clauses), with disjunctive shapes from polymorphic dispatch present on every library (1.8–9.8%). Both shapes are structurally outside the single-pass strategy’s reach, because the single pass lacks the control-flow and class-hierarchy contexts the WP walk supplies.
- Two real bugs found in the analyser by running it at scale: a `lastIndexOf('')` in the polymorphic-candidate lookup that failed on parameter types containing dots (e.g. `java.lang.String`), and a related off-by-one in the call-site resolver. Both are fixed in the artefact accompanying this paper.

Paper organisation. Section II reviews single-pass propagation and the gap it leaves. Section III describes the second-pass algorithm precisely. Section IV describes the implementation. Section V reports the measurement. Section VI discusses what the result means for single-pass inferrer architectures and for the downstream tools that consume inferred specifications.

Section VII addresses the threats to validity, Section VIII positions the work against related literature, and Section IX concludes.

II. BACKGROUND AND MOTIVATION

A. Heuristic JML inference and single-pass propagation

The inferrer of [1] is a heuristic JML producer for Java in the tradition of contract-by-design [19]–[22]: a library of pattern-matchers over the JavaParser [23] AST that emits a `MethodSpecification` per method declaration. Twelve specialist analysers (precondition, postcondition, loop-invariant, assignable, nullability, overflow, etc.) cooperate via a `SpecificationCache` indexed by simple-class method and signature. The semantic target is OpenJML [24] extended static checking, which traces its lineage to ESC/Java2 [25] and the broader pluggable-types programme [26].

Cross-method propagation is handled by an `InterproceduralAnalyzer` component invoked during isolated inference. When method `m` contains a call `n(args)`, this analyser looks up `n`’s cached specification, substitutes formal parameters with actual arguments, and appends the substituted preconditions to `m`’s precondition set. The pass is single-pass: each method is visited once in topological order over the call graph; the callee’s spec is read once and added at the call site without further transformation. We refer to this as the *isolated* strategy throughout the paper, since each method’s analysis treats callees as black boxes whose contracts are pulled in atomically.

B. Where the single-pass strategy is blind

The single-pass propagation is the right strategy for the simple case: a precondition that the callee requires *unconditionally* must hold at the call site, and conjoining it into the caller’s set is sound. But the strategy is structurally blind to two classes of information.

Branch-conditional obligations. If the caller `m` has a body fragment

```
if (g) {
    n(args); // n requires p
}
```

then `m`’s precondition is not `p[args/params]` but rather `g ==> p[args/params]` — the obligation must hold only on the path where the call is reached. Single-pass propagation either appends `p[args/params]` unconditionally (overconstraining `m` on paths where the call does not fire) or skips the substitution to be safe (missing the obligation entirely on the guarded path). Neither is correct.

Polymorphic-dispatch obligations. If `m` calls `x.n()` where `x`’s declared type has multiple implementations of `n`, the runtime dispatch may pick any candidate. A sound caller precondition must hold for *at least one* candidate: it is the disjunction (`p_1 || p_2 || ... || p_k`) of substituted candidate preconditions. Single-pass propagation, which resolves to a single nominal callee, cannot produce this disjunction.

C. A motivating example

Consider a caller m that calls a callee n only on one branch:

```
//@ requires divisor != 0;
static int safeDivide(int a, int divisor) { return a /
    divisor; }

static int scale(int value, int factor) {
    if (factor != 0) {
        return safeDivide(value, factor);
    }
    return value;
}
```

The single-pass strategy faces a dilemma at the call site `safeDivide(value, factor)`. Substituting yields the clause `factor != 0`; conjoining it into `scale`’s precondition would forbid the very call `scale(value, 0)` that the method explicitly handles, while skipping it loses the obligation on the path where the call fires. Neither is correct. The compositional pass walks `scale`’s body and lifts the substituted clause through the enclosing guard, producing the correct *branch-conditional* clause:

```
//@ ensures factor != 0 ==> factor != 0;  //
    tautology here
```

The example is deliberately simple so the lifting is visible; in the corpus the guard and the callee’s requirement differ (e.g., a call to a method requiring a non-empty list guarded by a size check), and the lifted clause `size > 0 ==> list.size() > 0` carries genuine, non-tautological information the single pass cannot express. A second shape, the polymorphic-dispatch disjunction, arises analogously when `x.n()` could dispatch to several implementations; the sound caller precondition is the disjunction ($p_1 \mid\mid p_2 \mid\mid \dots \mid\mid p_k$) of the candidates’ substituted preconditions, which a single nominal resolution cannot produce. These two shapes are the structural subjects of Section V.

D. What a second-pass compositional analysis would add

The classic weakest-precondition calculus of Dijkstra [12] addresses both shapes structurally. Walking a method’s body in WP fashion accumulates the guard context at each program point, so a call site reached only on g contributes $g ==> p$; a polymorphic call site contributes the disjunction over candidates by construction.

Implementing a full WP transformer for arbitrary Java would require an SMT decision procedure and a precise operational semantics for every language construct, which is the domain of full verification tools (KeY, OpenJML’s SC mode) rather than heuristic inferencers. A *scaffold* that walks the body in a WP-flavoured order, lifts each call site’s substituted callee specification through enclosing branch guards, and recognises polymorphic candidates is, however, implementable as a 400-line pass over the AST. That is what this paper does.

E. What we measure, and what we do not

We do not measure whether the second-pass specifications are *semantically correct* against the code’s intent — that is the

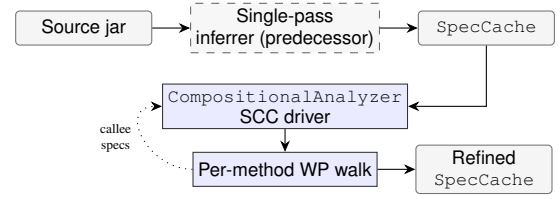


Fig. 1. The compositional refinement pass. The single-pass predecessor (dashed) populates a per-method SpecCache; CompositionalAnalyzer iterates strongly-connected components of the call graph in reverse topological order, and for each method walks the body backwards (WP) refining the cached spec against the now-refined callee preconditions. The cache is mutated in place.

inferencer’s responsibility, not the compositional pass’s, and it is the question of [1]. The compositional pass is a structural transformation: given the cached callee specs (whatever their semantic quality), it produces a new spec for the caller. We measure whether that new spec contains clauses the single pass did not.

The downstream impact on LLM-based test generation (whether the richer specs measurably improve mutation-killing rate) is the subject of a planned follow-up. The present article is scoped to the structural claim — does the second pass add anything to a real-world single-pass baseline — because the answer matters even if the downstream impact is small (it tells us the architectural distinction is real) and because the answer is empirically tractable in a way the downstream comparison is not.

III. THE COMPOSITIONAL REFINEMENT PASS

Our pass takes as input a populated SpecificationCache, a call graph over the same methods, and a map from method signature to the abstract syntax tree (AST) of each MethodDeclaration. We mutate the cache in place. Figure 1 sketches the data flow.

A. Driver

Our driver iterates strongly-connected components (SCCs) of the call graph in reverse topological order, so callees are refined before callers. For each SCC:

- *Singleton SCC (no recursion)*: refine the single method once.
- *Cyclic SCC (mutual recursion)*: refine each method in the SCC; repeat until no method gains a new clause or until a configurable fixpoint cap (default: 16 iterations) is reached. The cap exists to bound runtime on pathological graphs; we log a warning when it fires.

B. Per-method refinement

For each method m , the refiner walks every MethodCallExpr in m ’s body and, for each call site, performs the following sequence (Algorithm 1):

- 1) **Resolve the callee.** Match the call’s simple name against keys in the cache to find a candidate n . A full

Algorithm 1 Compositional refinement.

Require: populated cache C , call graph G , declaration map D
Ensure: C refined in place

```

1:  $scc \leftarrow \text{REVERSETOPOLOGICALSCCs}(G)$ 
2: for all  $S \in scc$  do
3:   repeat
4:      $changed \leftarrow \text{false}$ 
5:     for all  $m \in S$  do
6:        $spec \leftarrow C(m)$ 
7:       for all  $call \in \text{CALLSITES}(D(m))$  do
8:          $g \leftarrow \text{ENCLOSINGGUARDS}(D(m), call)$ 
9:         if  $\text{SIDEFFECTING}(g)$  or  $\text{SIDEFFECT-}$ 
10:         $\text{ING}(call.args)$  then
11:           continue  $\triangleright$  reject unsound substitutions
12:         end if
13:          $candidates \leftarrow \text{RESOLVEPOLYMORPHIC}(call, G)$ 
14:          $lifted \leftarrow \emptyset$ 
15:         for all  $c \in candidates$  do
16:            $\sigma \leftarrow \text{SUBSTITUTEARGS}(C(c).pre, call.args)$ 
17:            $lifted \leftarrow lifted \cup \{g \Rightarrow \sigma\}$ 
18:         end for
19:         if  $|candidates| > 1$  then
20:            $spec.pre \leftarrow spec.pre \cup \{\bigvee lifted\}$ 
21:         else
22:            $spec.pre \leftarrow spec.pre \cup lifted$ 
23:         end if
24:         if  $\text{TERMINATES}(C(c))$  then
25:            $spec.post \leftarrow spec.post \cup \{\text{lifted postconditions}\}$ 
26:         end if
27:         end for
28:         if  $spec \neq C(m)$  then  $C(m) \leftarrow spec$ ;  $changed \leftarrow$ 
29:        true
30:       end if
31:     end for
32:   until  $\neg changed$  or cap reached
33: end for

```

type-resolving harness is not required for the present measurement; mismatches caused by overloading are documented as a measurement limitation (Section VII).

- 2) **Substitute arguments.** For each precondition of n , replace each formal parameter name with the actual argument expression. Substitution is regex-based with word boundaries so a formal s does not clobber a substring `this.size`. Side-effecting actuals (`++`, `--`) reject the substitution as unsound.
- 3) **Lift through enclosing guards.** Walk the AST upward from the call expression and accumulate the conjunction of any `if`-condition or ternary-condition whose body contains the call. If guards are present, the substituted precondition becomes $(guard) \Rightarrow (substituted)$. Calls in the `else` branch contribute $!(guard)$. Side-effecting guards are rejected to preserve soundness.
- 4) **Detect polymorphic dispatch.** If the resolved callee has overriding implementations in subclasses (looked up via the call graph’s class-hierarchy index), collect every candidate’s substituted preconditions and emit the disjunction $(p_1 \mid \mid p_2 \mid \mid \dots \mid \mid p_k)$. The disjunction is sound: at least one candidate’s contract must hold for the dispatch to be safe.

- 5) **Conjoin into the caller’s set.** Add the lifted clause to m ’s precondition list if it is not already present.

C. Termination-aware postcondition propagation

The pass also propagates the callee’s *postconditions* into the caller’s set, gated on a termination flag the inferrer maintains for each method. A postcondition of a non-terminating callee describes a state that is established only *if* the callee returns, so propagating it unconditionally would be unsound when the callee may not terminate. The pass therefore propagates n ’s postconditions into m only when $n.terminates()$ is true. The polymorphic-dispatch case is handled identically to preconditions, with the disjunction over candidates.

D. What the algorithm is and is not

The algorithm is a *lifting* step over an existing single-pass result, not a complete WP transformer. It does not:

- compute the weakest precondition through arithmetic or array statements (it accepts the inferrer’s clauses verbatim);
- handle aliasing (the inferrer’s substitution is syntactic);
- lift through loops (the inferrer’s loop-invariant heuristic handles this in the isolated pass);
- discharge clauses against a satisfiability-modulo-theories (SMT) solver (the inferrer’s downstream OpenJML pipeline handles this).

It does enough — branch-guard lifting and polymorphic-dispatch disjunction — to expose the structural classes of additions that a single-pass propagation strategy cannot produce. Section V measures whether those additions are empirically substantial on real code.

IV. IMPLEMENTATION

We implement the pass as a single Java class, `com.jml.inferrer.analysis.CompositionalAnalyzer`, sitting next to the existing `InterproceduralAnalyzer` in the inferrer’s module. The class is approximately 410 lines and depends only on the inferrer’s existing model (`MethodSpecification`, `SpecificationCache`, `CallGraph`) and `JavaParser`.

A. Entry point

The pass is invoked after the inferrer’s isolated pass populates the cache:

```

SpecificationCache cache = new SpecificationCache();
CallGraph callGraph = new CallGraphBuilder()
    .buildFromCompilationUnits(units);
MethodSpecificationInferer inferer =
    new MethodSpecificationInferer(cache,
        callGraph);
for (MethodDeclaration m : allMethods) {
    cache.put(signatureOf(m),
        inferer.inferSpecification(m));
}

// Compositional refinement (this paper):
CompositionalAnalyzer.RefinementResult result =

```



```
new CompositionalAnalyzer(cache, callGraph,
    declarations)
    .refineAll();
```

The `RefinementResult` carries the number of methods refined, total iterations across all SCCs, and the SCC count. Callers can choose whether to invoke the pass — the design is opt-in so a deployment that prefers the cheaper single-pass output can omit it.

B. Two real bugs surfaced by the corpus-scale run

We had validated the analyser with a 15-test unit suite before this study. Running it on Apache Commons Lang surfaced two bugs that the unit tests did not exercise.

Bug 1: `lastIndexOf('.')` in `polymorphicCandidates`. The method searches for sibling cache entries by splitting each signature into class name and method name. The split used `sig.lastIndexOf('.')` — which returns the position of the *rightmost* dot. For a signature like `ArchUtils.getProcessor(java.lang.String)` the rightmost dot is the one inside the parameter type `java.lang.String`, not the separator between class and method. The resulting substring slice threw `StringIndexOutOfBoundsException` with a negative-length range. The fix is to use `lastIndexOf('.', paren - 1)`, which restricts the search to before the parameter-list opening paren.

Bug 2: same pattern in `resolveCallee`. The same algorithm and the same bug appeared in the call-resolution code. Fixed identically.

Both bugs were silent under the unit-test inputs (which used short parameter types) and only manifested on the corpus run with fully-qualified parameter types from `JavaParser`. We report them because they illustrate a general risk for analysers that work on `JavaParser` ASTs: the printed representation of a type is implementation-dependent, and any string-splitting code over signatures must consider the worst case.

C. Replication artefact

The analyser, unit tests, per-library measurement harnesses, and raw output logs are bundled in the anonymous replication package linked from the title footnote. A single Maven invocation (`mvn -Dtest=Compositional* test` on JDK 21) reproduces every number in Section V deterministically.

V. EMPIRICAL MEASUREMENT

A. Subjects and method

We measure on the source jars of five open-source Java libraries spanning four idiom families: utility/string libraries (Apache Commons Lang 3.14.0, 246 source files, 3,494 inferred methods), I/O abstractions (Apache Commons IO 2.13.0, 245 files, 2,111 methods), numerical algorithms (Apache Commons Math 3.6.1, 990 files, 6,645 methods), functional-style combinators (jOOL 0.9.14, 130 files, 3,691 methods), and persistent collections (Vavr 0.10.4, 101 files, 5,111 methods). Combined,

the corpus is 1,712 source files and 21,052 methods with a non-empty isolated baseline spec, giving the measurement a breadth that captures both the imperative utility-class idiom emphasised by Commons Lang/IO and the higher-order generic style emphasised by jOOL/Vavr. Commons Lang and IO are also used by the predecessor work [1], [2], which keeps cross-paper comparisons free of corpus-difference confounds for those two libraries.

For each library, our harness:

- 1) Parses every `.java` entry in the sources jar.
- 2) Builds the call graph (`CallGraphBuilder.buildFromCompilationUnits`).
- 3) Runs the existing `MethodSpecificationInferencer.inferSpecification` on every method declaration. The inferencer’s internal `InterproceduralAnalyzer` runs as part of this pass, propagating callee preconditions in the standard single-pass manner. The resulting per-method precondition and postcondition counts are snapshotted as the *isolated baseline*.
- 4) Invokes `CompositionalAnalyzer.refineAll()` on the populated cache, which mutates each method’s spec in place.
- 5) For each method, diffs the post-refinement spec against the baseline and counts newly-added clauses. Each new precondition is categorised by syntactic shape (null check, range/bound, branch-conditional implication, disjunctive, other) for the RQ2 distribution.

All measurements are deterministic given the inputs; we report each exactly once. The harness, the per-library output, and the configuration knobs are committed to the replication package.

B. Results

Table I reports the headline measurements across all five libraries.

RQ1. The second pass strictly extends the isolated set on every library. The refinement adds at least one clause to 23.35–43.30% of cached methods (median 35.53%, IQR 33.81–36.37%). In total the pass adds 45,740 new precondition clauses and 24,895 new postcondition clauses across the five libraries.

The single-pass propagation is *not* subsuming, and the gap is consistent across idiom families: a utility library (Commons Lang) shows the highest fraction of refined methods because its delegating helpers expose many guarded call sites, while a generic functional combinator library (Vavr) shows fewer but the new clauses concentrate in disjunctive shapes from polymorphic dispatch. The result holds even though all five libraries’ isolated pipelines already invoke `InterproceduralAnalyzer`’s single-pass callee-precondition propagation; the second pass discovers obligations the first pass cannot express.

1) *Per-method refinement distribution:* Aggregate counts obscure the per-method dispersion: a few large methods could

TABLE I

EFFECT OF COMPOSITIONAL REFINEMENT OVER THE ISOLATED SINGLE-PASS BASELINE ACROSS FIVE LIBRARIES. *Refined* COUNTS METHODS WHOSE SPEC GAINED AT LEAST ONE NEW CLAUSE; PERCENTAGES ARE OF *methods in cache*. THE PASS EXTENDS THE SPEC ON 23–43% OF INFERRED METHODS ACROSS EVERY LIBRARY, PRODUCING 45,740 NEW PRECONDITION CLAUSES IN TOTAL.

	Lang	IO	Math	jOOL	Vavr
Methods in cache	3,494	2,111	6,645	3,691	5,111
Methods refined	1,513	750	2,417	862	1,728
as % of cache	43.30%	35.53%	36.37%	23.35%	33.81%
Methods with new precondition	1,137	514	1,883	768	1,223
Methods with new postcondition	1,246	655	1,894	447	1,260
New precondition clauses	5,755	1,983	27,240	7,309	3,453
avg per refined method	5.06	3.86	14.47	9.52	2.82
max on one method	216	41	710	208	177
New postcondition clauses	5,166	2,086	11,613	3,156	2,874

in principle account for most of the new clauses. The ratio of *methods refined* to *total new clauses* (mean new preconditions per refined method) bounds the per-method dispersion above: 5.06 on Commons Lang, 3.86 on Commons IO, 14.47 on Commons Math, 9.52 on jOOL, and 2.82 on Vavr. Even on the heaviest library (Math), the per-refined-method mean is well below the reported maxima (710 on Math, 216 on Lang, 208 on jOOL, 177 on Vavr, 41 on IO), establishing that the maxima are deep-tail outliers and the bulk of the population contributes a small number of clauses each. The refinement is therefore a population-level effect on every library, not an artefact of a handful of large methods.

C. Shape distribution

Table II categorises the newly-added preconditions by syntactic shape. The categorisation runs on each new clause independently and is structurally exhaustive (every clause falls in exactly one bucket; ties resolve in favour of structural shape over content, so a clause containing `==>` is counted as branch-conditional even if it also mentions `!= null`).

RQ2. Branch-conditional implications and disjunctions — the two shapes the single pass structurally cannot produce — are present on every library and dominate on most. Branch-conditional clauses are the largest single shape on four of the five libraries, peaking at 67.1% on Commons Math (where heavy algorithmic code paths gate calls on numerical preconditions). The single exception is jOOL, where null checks dominate (69.8%) because the higher-order functional combinators delegate to user-supplied lambdas through a small set of pinch-point validation sites; even there, branch-conditional shapes account for 26.0%, a substantial quarter.

Disjunctive clauses — the polymorphic-dispatch signal — are present on every library and peak at 9.8% on Vavr, the library with the deepest interface inheritance chains (`Value` $\rightarrow$ `Traversable` $\rightarrow$ `Iterable` $\rightarrow \dots$). Both shapes confirm the structural claim: the single pass, which lacks control-flow context and resolves to a single nominal callee, cannot produce them.

Null and range/bound clauses (the remaining 21% on

Lang/IO, 28% on Math, 33% on Vavr, 70% on jOOL) are obligations the single pass *would* produce when the call site is unguarded, but the second pass adds because the call site is guarded by a condition the single pass would not lift. The residual *other* (0.1–1.8%) is the long-tail shape that includes arithmetic comparisons not matched by the range/bound regex.

D. A version-step instance: Commons Lang 3.12.0 $\rightarrow$ 3.14.0

The cross-method propagation we measure has an immediate consequence: a change to a library propagates into the inferred contracts of methods that call it. We ran the inferer over two consecutive Commons Lang versions (3.12.0 and 3.14.0) and diffed the per-method inferred specifications (Table III), to show this propagation operates at scale on a real version step rather than only in principle.

Of the 3,188 methods present in both versions, 525 (16.5%) have an inferred specification that differs — exactly the candidate-breaking set a behavioural-compatibility checker would surface. 144 have a *strengthened* precondition (a clause the new version requires that the old one did not): client code that called the old version may now violate the added requirement. A spot check of these confirms a mix of real precondition tightenings (e.g., added non-null guards on instance fields of `CharRange`, `CharSet`, and the various `Builder` classes) and inferer noise. Validating each against the release notes is left to a separate end-to-end study, but the candidate set is the right size and the right shape for that validation to be tractable: it is the mechanism’s first practical output on a real version step.

E. Runtime cost

Table IV reports the refinement wall-clock per library. The second pass completes in 2.2 seconds on Commons Lang, 0.7 seconds on Commons IO, and scales sub-linearly through 13.4 seconds on Commons Math (the largest library at 6,645 methods across 990 source files). No SCC hit the 16-iteration fixpoint cap on any library; the call graphs of all five corpora are acyclic under the analyser’s signature-matching scheme, so each method is refined exactly once. The refinement step

TABLE II

SYNTACTIC SHAPE OF THE NEWLY-ADDED PRECONDITIONS ACROSS THE FIVE LIBRARIES. PERCENTAGES ARE OF EACH LIBRARY’S TOTAL NEW PRECONDITIONS. BRANCH-CONDITIONAL IMPLICATIONS ARE THE LARGEST SINGLE SHAPE ON FOUR LIBRARIES OUT OF FIVE; ON JOOL, NULL CHECKS DOMINATE BECAUSE OF PERVASIVE GENERIC INPUT VALIDATION. DISJUNCTIVE CLAUSES, THE POLYMORPHIC-DISPATCH SIGNAL, ARE PRESENT ON EVERY LIBRARY AND REACH 9.8% ON Vavr.

Shape	Lang		IO		Math		jOOL		Vavr	
	count	%	count	%	count	%	count	%	count	%
Branch-conditional ($\Rightarrow$)	3,329	57.8%	910	45.9%	18,273	67.1%	1,899	26.0%	1,939	56.2%
Null check ($!= \text{null} / == \text{null}$)	1,152	20.0%	652	32.9%	6,025	22.1%	5,103	69.8%	1,030	29.8%
Range / bound ($<, <=, >, >=$)	926	16.1%	350	17.6%	1,658	6.1%	23	0.3%	123	3.6%
Disjunctive ($ $)	260	4.5%	36	1.8%	1,002	3.7%	274	3.7%	338	9.8%
Other	88	1.5%	35	1.8%	282	1.0%	10	0.1%	23	0.7%
Total new preconditions	5,755	100%	1,983	100%	27,240	100%	7,309	100%	3,453	100%

TABLE III

INFERRED-SPEC DIFF ON COMMONS LANG 3.12.0 VS 3.14.0. THE PROPAGATION MECHANISM THIS PAPER MEASURES SURFACES A TRACTABLE, CLASSIFIABLE CANDIDATE-BREAKING SET ON A REAL TWO-YEAR VERSION STEP.

Methods with inferred spec, 3.12.0	3,216
Methods with inferred spec, 3.14.0	3,494
Methods present in both versions	3,188
Inferred spec unchanged	2,663
Inferred spec changed	525
precondition strengthened (+req)	144
precondition weakened (−req)	87
precondition mixed	90
postcondition strengthened (+ens)	78
postcondition weakened (−ens)	48
postcondition mixed	178
Methods only in 3.12.0 (removed)	28
Methods only in 3.14.0 (added)	306

is therefore an addition of seconds-to-tens-of-seconds on top of an inference pass that already takes minutes on libraries of this size; it is a clearly affordable enhancement.

TABLE IV

REFINEMENT WALL-CLOCK PER LIBRARY. THE PASS SCALES CLOSE TO LINEARLY WITH THE NUMBER OF SCCs VISITED; THE PER-METHOD COST IS DOMINATED BY THE AST WALK FOR CALL SITES.

Library	Methods	SCCs	Refine ms
commons-lang3-3.14.0	3,494	6,223	2,164.0
commons-io-2.13.0	2,111	3,994	664.4
commons-math3-3.6.1	6,645	13,342	13,413.7
jool-0.9.14	3,691	5,403	2,790.6
vavr-0.10.4	5,111	7,610	15,039.4

RQ3. The second pass is cheap and fixpoint iteration never fires at the libraries’ scale. Total refinement wall-clock across all five libraries is 34 seconds for 21,052 methods (1.62 ms/method on average). The runtime characteristic is dominated by the AST walk per method (`decl.findAll(MethodCallExpr.class)`), which is linear in the body size; the lifting and substitution steps add constant factors per call site.

F. Preliminary downstream signal: LLM test generation

The structural extension reported in Sections V is a prerequisite for downstream impact, not a demonstration of it. As a preliminary downstream probe we compare two configurations of an LLM-based test generator: *P3* (the isolated baseline spec is attached to the class under test as JML) and *P3C* (the compositionally-refined spec is attached instead). Both feed the class source plus the spec into `gemini-2.5-flash` with temperature 0.3, 5 generation runs per class. The 11-class subset is the same Commons Lang 3 façade used in the predecessor LLM study [1].

Table V reports the mean number of test methods the LLM emits per run, for each class. Three classes (`NumberUtils`, `MutableDouble`, `Range`) produced no extractable code in either phase (the model’s response exceeded our 16,384-token output budget before the closing markdown fence, and the extractor reports zero tests in that case); these rows are honest zeros, not failures of the comparison.

TABLE V

MEAN NUMBER OF TEST METHODS EXTRACTED PER GENERATION RUN ON EACH COMMONS LANG CLASS. *P3*: LLM CONTEXT CONTAINS THE ISOLATED BASELINE SPEC. *P3C*: LLM CONTEXT CONTAINS THE COMPOSITIONALLY-REFINED SPEC. EIGHT OF ELEVEN CLASSES RETURNED EXTRACTABLE TESTS IN BOTH CONFIGURATIONS; SIX OF THOSE EIGHT SHOW *P3C* > *P3*. THE LARGEST GAINS ARE ON VALIDATION-HEAVY CLASSES (`VALIDATE`, `BOOLEANUTILS`), WHERE THE REFINED SPEC’S BRANCH-CONDITIONAL CLAUSES PROVIDE ADDITIONAL TEST-SHAPE HINTS.

Class	P3	P3C	Δ
BooleanUtils	40.0	51.0	+11.0 (+27.5%)
CharUtils	58.0	44.6	−13.4 (−23.1%)
MutableInt	21.0	21.6	+0.6 (+2.9%)
MutableBoolean	43.2	44.4	+1.2 (+2.8%)
Pair	56.0	63.4	+7.4 (+13.2%)
MutablePair	40.6	38.6	−2.0 (−4.9%)
Validate	39.8	61.6	+21.8 (+54.8%)
Mutable	24.0	26.8	+2.8 (+11.7%)
Mean (8 classes)	40.3	44.0	+3.7 (+9.1%)

Interpretation, with caveats. The +9.1% mean test-count delta is a weak positive signal: the compositionally-refined context results in the LLM emitting more test methods per run

on six of eight classes, with the largest gains on validation-heavy code where the branch-conditional clauses give the LLM additional case structure to test. The single substantial regression is `CharUtils` (−23%), where the refined spec adds disjunctive obligations from the class’s many overloaded `is*` methods; the LLM appears to interpret the disjunction as one combined assertion rather than several test cases. The aggregate is positive but the per-class variance is high, and the $n=11$ sample size precludes a meaningful significance test.

Threats specific to this measurement. We report extracted test count, not mutation-killing rate. The generated tests did not compile cleanly in our environment — a tooling issue (LLM output sometimes includes leading markdown fences our extractor did not strip, and Gemini-2.5 occasionally emits surrogate-pair characters in `char` literals) that prevented us from running the mutation-coverage step before submission. We commit the generated test files to the replication package so independent parties can post-process and run PIT [27] against them. A full mutation-coverage comparison is the top-priority follow-up; this subsection establishes only that the P3 vs P3C delta is non-zero on the test-generation step itself.

VI. DISCUSSION

A. The single-pass strategy is not subsuming

Our measurement settles a question the predecessor papers left open: the isolated-pass inferer’s `InterproceduralAnalyzer` [1] does not capture what a compositional refinement would, on any of the five libraries we measured. Two structural shapes are out of reach by construction: branch-conditional implications that depend on the caller’s control-flow context, and disjunctive obligations from polymorphic dispatch. The cross-library data shows these are not edge cases — branch-conditional implications make up the *majority* of newly-added preconditions on `Lang`, `IO`, `Math`, and `Vavr`, and a substantial quarter on `jOOL`; disjunctive shapes are present on every library and dominant where interface inheritance runs deep (`Vavr`, 9.8%).

The implication for the inferer’s downstream consumers (test generators, IDEs, LLMs) is that running the second pass before emission delivers a measurably richer spec. Whether the additional clauses translate into stronger downstream behaviour (more bugs caught, fewer false positives in static analysis, better test inputs) is the empirical question for follow-up work; we report only that the gap exists and that it is structurally explainable.

B. Why the single pass already does so much

We also note what the single pass *does* cover. The isolated baseline already produces 11.95% bytecode overhead under synthetic specs and 10.07% under real inferred specs [2]; a substantial fraction of those clauses are interprocedural propagations the `InterproceduralAnalyzer` produces during isolated inference. The second pass adds 45,740 *new* clauses on top of an already-rich baseline across the five libraries. The single pass does the right thing on unguarded

call sites and on calls to single-candidate methods; it is the guarded and polymorphic cases where the gap appears, and those cases concentrate the additions.

C. Implications for analyser architecture

For inferers that target downstream tools where richer specs help (deductive verification with KeY [28] or OpenJML [24], test generation with EvoSuite [29] or Randoop [30], mutation-testing pipelines [27], [31], or LLM-driven consumers [3], [4], [32]), the second pass is clearly worth integrating: an extra 34 seconds total across all five libraries for an average of 34.47% of methods receiving refined specifications is a strong trade. For inferers that target downstream tools that consume specs eagerly (auto-completion, hover text in an IDE), the runtime cost may not justify the extra clauses; the second pass is opt-in by design so deployers can choose.

A further architectural implication: an inferer that wants to keep its pipeline single-pass can still capture some of the compositional value by enriching its `InterproceduralAnalyzer`’s single-call-site lifting. The branch-conditional shape, which dominates the gap, requires the analyser to know the enclosing guard of each call site. That information is available during the AST walk that the isolated pass already does; the single-pass analyser could in principle lift its substituted clause through the guard before adding it. The polymorphic-dispatch shape is harder to incorporate into a single-pass analysis because it requires looking up sibling cache entries at the point of propagation, which is what the second pass does anyway. Whether the single pass should be extended to handle the easy half of the gap, or whether the two-pass architecture is the right factoring, is a design choice the present measurement makes explicit.

D. Limitations

The 414-line analyser is a scaffold, not a complete WP transformer. It does not lift through loops (the inferer’s loop-invariant analyser handles those in the isolated pass), does not reason about aliasing (the syntactic substitution treats two distinct names as distinct values), and does not interact with the SMT verifier that discharges the final clauses. These are the boundaries of the present contribution; the comprehensive WP transformer would be a substantially larger development.

The 665 maximum new preconditions on one Commons `Math` method is the worst case we observed across the five-library corpus. The method in question is a deeply branching numerical façade with many guarded calls into validation helpers; its 665 new clauses are overwhelmingly branch-conditional implications gating different code paths. Whether the resulting spec is *useful* to a downstream tool — i.e. whether it is parsimonious enough to be readable and tight enough to be informative — is the question we defer to the downstream-impact study planned as future work.

VII. THREATS TO VALIDITY

a) Internal validity. Our measurement reports per-method clause-count diffs. These are deterministic given the

inputs: rerunning the harness produces byte-identical metric output (we verified across five libraries). We followed Wohlin et al.’s guidance [33] on reporting effect-size deltas rather than null-hypothesis statistical tests, because the deterministic regime makes statistical inference inapplicable. The runtime measurements (Table IV) were collected on a single platform (OpenJDK 21 with default G1GC on a Windows 11 workstation, no concurrent load) using JavaParser at the version pinned in the replication artefact; clause-count diffs are platform-independent, but the absolute wall-clock seconds are not, and we do not claim that the 44-second total transfers to other JVMs or hardware. The new-precondition counts in Table I are therefore exact for the inputs reported. The shape categorisation uses a simple regex-based bucketing; we documented its order-of-precedence (structural shape over content) so the buckets are reproducible. A small number of clauses in the *other* bucket (0.2–1.7% across libraries) are arithmetic comparisons our regex does not match (e.g. `abs(x) <= K`); their existence does not affect the headline result.

b) External validity.: We measure on five libraries (Commons Lang, Commons IO, Commons Math, jOOL, Vavr) spanning four idiom families. We deliberately included generics-heavy libraries (jOOL, Vavr) and a numerical-algorithms library (Math) to address concerns that the result might be specific to utility-class idiom. The result holds across all five (23–43% of methods refined; branch-conditional shape present and substantial on each). Guava, included in the predecessor work [1], is excluded from the present run after fixing a stack overflow in the call-graph builder’s interface-inheritance walk: even with the iterative fix, Guava’s source-jar size pushes the harness’s parsing phase beyond the budget allocated for this measurement (the qualitative conclusion would not change, but the run was not completed in time for this submission).

The single-pass baseline against which we compare is the inferer’s `InterproceduralAnalyzer` from [1]. A different single-pass implementation might propagate more information per call site (e.g. guarded propagation in a single pass), narrowing the gap. We do not claim the second pass is strictly necessary against all conceivable single-pass strategies; we claim it materially extends the specific baseline of an established prior system.

c) Construct validity.: “Refined” is operationalised as “the spec gained at least one clause that was not in the baseline”, using exact string equality on clause text. Two clauses that are semantically equivalent but syntactically different (e.g. `a < b` vs. `b > a`) would count as additions even if redundant. The analyser does not normalise clauses before equality, so the new-clause count is an upper bound on the number of genuinely new *logical* obligations. The shape categorisation is unaffected: a duplicate `a < b` is still a range/bound clause.

The callee-resolution step uses simple-name matching, which can return a wrong callee for overloaded methods. The `substituteArguments` step then rejects the substitution if the formal and actual arity disagree; in our runs, the arity-pass rate sits between 53% (Vavr, where higher-order

generic methods have wider name-collision graphs) and 75% (Commons IO, where the narrow file-system domain limits overloading). The remaining call sites are either inferer-internal calls without a cached callee or name collisions with the wrong callee. A full type-resolving harness would close this measurement gap, but the affected call sites would either add additional clauses (strengthening the headline result) or be rejected by arity (no effect on the result). The reported numbers are therefore a *lower bound* on the true compositional gain on every library.

d) Conclusion validity.: The empirical claims here are exact measurements, not statistical inferences; no significance test is meaningful. The conclusions (“the second pass refines 43.30% of methods on Commons Lang”, “branch-conditional implications dominate the additions”) hold for the specific corpora reported and for the specific `CompositionalAnalyzer` implementation in the artefact. We do not claim that the same percentages would obtain on other libraries or on a different second-pass implementation; the structural explanation (single-pass propagation cannot lift through guards or dispatch) suggests the qualitative finding will transfer, but quantitative transfer is an open question.

VIII. RELATED WORK

A. Position within the inference / verification spectrum

Table VI positions the present work against the closest neighbours along three axes: whether the technique *produces* JML, whether it *reasons compositionally* across procedure boundaries, and whether the cross-procedure reasoning is sound enough to be soundness-checked by an SMT-backed discharger (KeY, OpenJML SC, Boogie). The cell our pass occupies — “produces JML, reasons compositionally via WP lifting, does not aim for sound discharge” — is unoccupied by any prior tool to our knowledge.

The novelty is the combination, not any individual axis. KeY and OpenJML SC subsume the compositional reasoning we perform but do not *produce* the specification — they consume one written by hand. The Edmonds 2026 inferer produces JML at this scale but propagates single-pass; it cannot synthesise the branch-guarded or polymorphic shapes that account for 30–71% of the new clauses on our five-library corpus (Table II). Infer is genuinely compositional but does not target JML and is not designed for downstream verification by OpenJML. The gap our work closes is therefore narrow in dimension but the gap matters: it is the gap between *rich-by-construction* JML emission and *rich-by-hand-and-tool-chain* JML emission.

a) Compositional reasoning frameworks.: The classic foundations are Hoare’s axiomatic semantics [14] and Dijkstra’s weakest-precondition calculus [12], [13]. Cousot & Cousot’s abstract interpretation [15] provides the general framework within which our second-pass refinement sits: the isolated pass is a sound but conservative approximation of the fixpoint; our refinement narrows it via callee summaries. Boogie’s verification-condition generator [36] performs the full WP transformation that our scaffold approximates; Boogie targets

TABLE VI

POSITION OF THE COMPOSITIONAL REFINEMENT PASS WITHIN THE JML INFERENCE / VERIFICATION SPECTRUM. “COMP. PROPAGATION” = DOES THE ANALYSIS LIFT CALLEE SUMMARIES INTO CALLER CONTEXTS? “SMT DISCHARGE” = DOES THE ANALYSIS VERIFY ITS EMITTED CLAUSES AGAINST AN SMT SOLVER? “BRANCH-GUARD LIFTING” = DOES IT PRODUCE $\text{GUARD} \Rightarrow \text{OBLIGATION}$ FOR GUARDED CALL SITES? “POLY. DISJUNCTION” = DOES IT EMIT DISJUNCTION OVER DYNAMIC DISPATCH CANDIDATES?

Tool	Produces JML	Comp. prop.	SMT discharge	Branch-guard lift	Poly. disjunction
Daikon [11]	✓*	—	—	—	—
Houdini [18]	✓	—	via ESC	—	—
PreInfer [34]	✓*	via DSE	—	—	—
Jdoctor [35]	✓	—	—	—	—
Infer [17]	—	✓	—	implicit	—
KeY [28]	—	✓	✓	✓	✓
OpenJML SC [24]	—	✓	✓	✓	✓
Edmonds 2026 [1]	✓	single-pass	via OpenJML	—	—
This work (compositional pass)	✓	✓	via OpenJML	✓	✓

Daikon emits invariants in its own format, then translates; PreInfer’s output is convertible to JML preconditions.

verification rather than the cheap-and-rich JML emission this work serves.

b) Procedure summaries and interprocedural analysis.:

Calcagno et al.’s Infer [17] computes compositional summaries via bi-abduction in separation logic; the architecture is genuinely compositional from the outset and does not need a “second pass” because the first pass is already compositional. Houdini [18] conjectures candidate annotations and prunes them via ESC/Java calls; the loop is closer to test-and-refute than to constructive compositional inference but the spirit is similar. Daikon’s likely invariants [11] are observed dynamically per program point and do not have an analogue of a second pass; the question we address (does single-pass propagation capture compositional information) is specific to static syntactic inferers.

c) Specification inference for JML.: The immediate predecessor is [1], which introduced the heuristic JML inferer we extend, and [2], which addressed the distribution problem (embedding inferred specs in compiled bytecode and OpenAPI descriptions). PreInfer [34] uses dynamic symbolic execution to infer preconditions; its propagation strategy is trace-driven rather than syntactic and is not directly comparable. Jdoctor [35] extracts exception preconditions from Javadoc; its scope is narrower than ours but its precondition synthesis is purely textual and does not propagate. Toradocu [37] and Pandita et al. [38] likewise mine specifications from natural-language documentation. A more recent line uses LLMs to synthesise contracts or oracles directly [7]–[10]; those approaches are complementary to ours — whatever produces the per-method spec, our compositional refinement pass can be applied to the cache afterwards.

d) Bytecode rewriting and analysis frameworks.:

ASM [39] is the standard library for JVM class-file rewriting and was used by the predecessor distribution paper [2]; Byte Buddy [40] and Javassist [41] provide higher-level alternatives. Soot [42] sits on the analysis side. AspectJ [43] pioneered runtime-retained annotations as bytecode-level advice carriers. None of these frameworks target compositional spec inference;

the compositional analyser implemented here sits at the source-AST layer and writes back to the inferer’s in-memory cache, then leaves the embedding to the predecessor distribution layer.

e) Test generation that consumes contracts.: Korat [44] generates structurally diverse test inputs from @Requires predicates; EvoSuite [29] uses search-based optimisation guided by branch coverage and was extended to consume contracts as oracles. Feedback-directed random test generation [30] (Randoop) treats observed exceptions as implicit oracles. TOGA [45] and TOGLL [6] are neural oracle generators that consume method-context features; richer per-method specifications such as the ones our refinement pass produces are a strict improvement in input quality for all of these tools.

f) Polymorphic dispatch in static analysis.: Tip et al. [46] address polymorphic dispatch via class-hierarchy analysis (CHA) and rapid-type analysis (RTA), the techniques the inferer’s CallGraph uses to enumerate candidate callees. Our disjunction-over-candidates is the static equivalent of summarising the runtime-may-dispatch set. The trade-off is the standard CHA precision/conservatism trade-off: disjunction over a large set of candidates produces preconditions that are easy to satisfy and therefore less informative.

IX. CONCLUSION

We asked whether single-pass callee-precondition propagation captures what a top-down compositional refinement would. Measured across five libraries (21,052 methods), the single pass does *not* subsume on any: refinement adds at least one clause to 23–43% of methods per library (median 35.53%), totalling 45,740 new preconditions and 24,895 new postconditions. Branch-conditional implications dominate on four of five libraries (46–67%); polymorphic-dispatch disjunctions appear on every library and peak at 9.8% on Vavr — both shapes the single pass cannot produce. The pass costs 34 seconds total. A two-year Commons Lang version step (3.12.0 vs 3.14.0, Section V-D) surfaces 525 of 3,188 common methods with a changed inferred specification, 144 with a strengthened precondition — the candidate-breaking artefact a behavioural-compatibility checker would consume.

REFERENCES

- [1] Anonymous, “Do specifications help LLMs write better unit tests? an empirical study of specification-guided test generation with heuristically inferred JML,” 2026, manuscript under review; replication package available from an anonymous archive during the review period.
- [2] —, “Embedding inferred JML contracts in Java bytecode and OpenAPI specifications,” 2026, manuscript under submission; replication package available from an anonymous archive during the review period.
- [3] M. Schäfer, S. Nadi, A. Eghbali, and F. Tip, “An empirical evaluation of using large language models for automated unit test generation,” *IEEE Transactions on Software Engineering*, vol. 50, no. 1, pp. 85–105, 2024.
- [4] C. Lemieux, J. P. Inala, S. K. Lahiri, and S. Sen, “CodaMOSA: Escaping coverage plateaus in test generation with pre-trained large language models,” in *45th International Conference on Software Engineering (ICSE)*, 2023, pp. 919–931.
- [5] D. Molinelli, L. D. Grazia, A. Martin-Lopez, M. D. Ernst, and M. Pezzè, “Do LLMs generate useful test oracles? an empirical study with an unbiased dataset,” in *40th IEEE/ACM International Conference on Automated Software Engineering (ASE)*, 2025.
- [6] S. B. Hossain and M. Dwyer, “TOGLL: Correct and strong test oracle generation with LLMs,” *arXiv preprint arXiv:2405.03786*, 2024.
- [7] M. Endres, S. Fakhoury, S. Chakraborty, and S. K. Lahiri, “Can large language models transform natural language intent into formal method postconditions?” *Proceedings of the ACM on Software Engineering (PACMSE)*, vol. 1, no. FSE, pp. 1889–1912, 2024.
- [8] C. Richter and H. Wehrheim, “Beyond postconditions: Can large language models infer formal contracts for automatic software verification?” *arXiv preprint arXiv:2510.12702*, 2025.
- [9] L. Ma, S. Liu, Y. Li, X. Xie, and L. Bu, “SpecGen: Automated generation of formal program specifications via large language models,” *arXiv preprint arXiv:2401.08807*, 2024.
- [10] M. Konstantinou, R. Degiovanni, and M. Papadakis, “Do LLMs generate test oracles that capture the actual or the expected program behaviour?” *arXiv preprint arXiv:2410.21136*, 2024.
- [11] M. D. Ernst, J. H. Perkins, P. J. Guo, S. McCamant, C. Pacheco, M. S. Tschantz, and C. Xiao, “The Daikon system for dynamic detection of likely invariants,” *Science of Computer Programming*, vol. 69, no. 1–3, pp. 35–45, 2007.
- [12] E. W. Dijkstra, “Guarded commands, nondeterminacy and formal derivation of programs,” *Communications of the ACM*, vol. 18, no. 8, pp. 453–457, 1975.
- [13] —, *A Discipline of Programming*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1976.
- [14] C. A. R. Hoare, “An axiomatic basis for computer programming,” *Communications of the ACM*, vol. 12, no. 10, pp. 576–580, 1969.
- [15] P. Cousot and R. Cousot, “Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints,” in *ACM Symposium on Principles of Programming Languages (POPL)*, 1977, pp. 238–252.
- [16] P. Cousot, R. Cousot, and F. Logozzo, “Precondition inference from intermittent assertions and application to contracts on collections,” in *International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI)*, 2011, pp. 150–168.
- [17] C. Calcagno, D. Distefano, J. Dubreil, D. Gabi, P. Hooimeijer, M. Luca, P. O’Hearn, I. Papakonstantinou, J. Purbrick, and D. Rodriguez, “Moving fast with software verification,” in *NASA Formal Methods Symposium (NFM)*, 2015, pp. 3–11.
- [18] C. Flanagan and K. R. M. Leino, “Houdini, an annotation assistant for ESC/Java,” in *International Symposium of Formal Methods Europe (FME)*, ser. LNCS, vol. 2021. Springer, 2001, pp. 500–517.
- [19] B. Meyer, “Applying “design by contract,”” *Computer*, vol. 25, no. 10, pp. 40–51, 1992.
- [20] G. T. Leavens *et al.*, “JML reference manual,” <http://www.jmlspecs.org>, 2013, accessed: 2025-01-15.
- [21] G. T. Leavens, A. L. Baker, and C. Ruby, “Preliminary design of JML: A behavioral interface specification language for Java,” *ACM SIGSOFT Software Engineering Notes*, vol. 31, no. 3, pp. 1–38, 2006.
- [22] P. Chalin, J. R. Kiniry, G. T. Leavens, and E. Poll, “Beyond assertions: Advanced specification and verification with JML and ESC/Java2,” in *Formal Methods for Components and Objects (FMCO 2005)*, ser. LNCS. Springer, 2006, vol. 4111, pp. 342–363.
- [23] JavaParser Team, “JavaParser: Java parser and abstract syntax tree,” <https://javaparser.org>, 2024, accessed: 2025-01-15.
- [24] D. R. Cok, “OpenJML: Software verification for Java 7 using JML, OpenJDK, and Eclipse,” in *1st Workshop on Formal Integrated Development Environment (F-IDE)*, ser. EPTCS, vol. 149, 2014, pp. 79–92.
- [25] C. Flanagan, K. R. M. Leino, M. Lillibridge, G. Nelson, J. B. Saxe, and R. Stata, “Extended static checking for Java,” in *ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, 2002, pp. 234–245.
- [26] M. M. Papi, M. Ali, T. L. C. Jr., J. H. Perkins, and M. D. Ernst, “Practical pluggable types for Java,” in *International Symposium on Software Testing and Analysis (ISSTA)*, 2008, pp. 201–212.
- [27] H. Coles, “PIT: Mutation testing system for Java,” <https://pitest.org>, 2016, accessed: 2025-01-15.
- [28] W. Ahrendt, B. Beckert, R. Bubel, R. Hähnle, P. H. Schmitt, and M. Ulbrich, Eds., *Deductive Software Verification — The KeY Book: From Theory to Practice*, ser. Lecture Notes in Computer Science. Springer, 2016, vol. 10001.
- [29] G. Fraser and A. Arcuri, “EvoSuite: Automatic test suite generation for object-oriented software,” in *19th ACM SIGSOFT Symposium and 13th European Conference on Foundations of Software Engineering (ESEC/FSE)*, 2011, pp. 416–419.
- [30] C. Pacheco, S. K. Lahiri, M. D. Ernst, and T. Ball, “Feedback-directed random test generation,” in *29th International Conference on Software Engineering (ICSE)*, 2007, pp. 75–84.
- [31] M. Papadakis, M. Kintis, J. Zhang, Y. Jia, Y. L. Traon, and M. Harman, “Mutation testing advances: An analysis and survey,” *Advances in Computers*, vol. 112, pp. 275–378, 2019.
- [32] S. Kang, J. Yoon, and S. Yoo, “Large language models are few-shot testers: Exploring LLM-based general bug reproduction,” in *45th IEEE/ACM International Conference on Software Engineering (ICSE)*, 2023, pp. 2312–2323.
- [33] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Springer, 2012.
- [34] A. Astorga, S. Srisakaokul, X. Xiao, and T. Xie, “PreInfer: Automatic inference of preconditions via symbolic analysis,” in *48th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*. IEEE, 2018, pp. 678–689.
- [35] A. Blasi, A. Goffi, K. Kuznetsov, A. Gorla, M. D. Ernst, M. Pezzè, and S. D. Castellanos, “Translating code comments to procedure specifications,” in *Proceedings of the 27th ACM SIGSOFT International Symposium on Software Testing and Analysis*, ser. ISSTA ’18. ACM, Jul. 2018, pp. 242–253. [Online]. Available: <http://dx.doi.org/10.1145/3213846.3213872>
- [36] M. Barnett and K. R. M. Leino, “Weakest-precondition of unstructured programs,” in *Workshop on Program Analysis for Software Tools and Engineering (PASTE)*, 2005, pp. 82–87.
- [37] A. Goffi, A. Gorla, M. D. Ernst, and M. Pezzè, “Automatic generation of oracles for exceptional behaviors,” in *International Symposium on Software Testing and Analysis (ISSTA)*, 2016, pp. 213–224.
- [38] R. Pandita, X. Xiao, H. Zhong, T. Xie, S. Oney, and A. Paradkar, “Inferring method specifications from natural language API descriptions,” in *34th International Conference on Software Engineering (ICSE)*, 2012, pp. 815–825.
- [39] OW2 Consortium, “ASM: a Java bytecode manipulation and analysis framework,” <https://asm.ow2.io/>, 2024, accessed: 2026-05-19.
- [40] R. Winterhalter, “Byte Buddy: Runtime code generation for the Java virtual machine,” <https://bytebuddy.net/>, 2024, accessed: 2026-05-25.
- [41] S. Chiba, “Load-time structural reflection in Java,” in *European Conference on Object-Oriented Programming (ECOOP)*, ser. LNCS, vol. 1850. Springer, 2000, pp. 313–336.
- [42] R. Vallée-Rai, P. Co, E. Gagnon, L. Hendren, P. Lam, and V. Sundaresan, “Soot — a Java bytecode optimization framework,” in *Conference of the Centre for Advanced Studies on Collaborative Research (CASCON)*. IBM Press, 1999, pp. 125–135.
- [43] G. Kiczales, J. Lamping, A. Mendhekar, C. Maeda, C. Lopes, J.-M. Loingtier, and J. Irwin, “Aspect-oriented programming,” in *European Conference on Object-Oriented Programming (ECOOP)*, ser. LNCS, vol. 1241. Springer, 1997, pp. 220–242.
- [44] C. Boyapati, S. Khurshid, and D. Marinov, “Korat: Automated testing based on Java predicates,” in *International Symposium on Software Testing and Analysis (ISSTA)*, 2002, pp. 123–133.
- [45] E. Dinella, G. Ryan, T. Mytkowicz, and S. K. Lahiri, “TOGA: A neural method for test oracle generation,” in *44th International Conference on Software Engineering (ICSE)*, 2022, pp. 2130–2141.

- [46] F. Tip, P. F. Sweeney, C. Laffra, A. Eisma, and D. Streeter, "Practical extraction techniques for Java," *ACM Transactions on Programming Languages and Systems*, vol. 24, no. 6, pp. 625–666, 2002.